10. Morphometric Characters

Characteristics describing the interactions of these elements, and the elements themselves are calculated at three scales: small - covering only aspects of the element; medium - covering aspects of the element and neighbouring elements and large - covering neighbouring elements up to five topological neighbours.

1. **Area of a building** is denoted as

$$a_{blg} \qquad (1)$$

and defined as an area covered by a building footprint in m^2 .

2. **Perimeter of a building** is denoted as

$$p_{blg} \qquad (2)$$

and defined as the sum of lengths of the building exterior walls in m.

3. **Courtyard area of a building** is denoted as

$$a_{blg_c} \qquad (3)$$

and defined as the sum of areas of interior holes in footprint polygons in m^2.

4. **Circular compactness of a building** is denoted as

$$CCo_{blg} = \frac{a_{blg}}{a_{blgC}} \qquad (4)$$

where a_{blgC} is an area of minimal enclosing circle. It captures the relation of building footprint shape to its minimal enclosing circle, illustrating the similarity of shape and circle (Dibble et al. 2015).

5. **Corners of a building** is denoted as

$$Cor_{blg} = \sum_{i=1}^{n} c_{blg} \qquad (5)$$

where c_{blg} is defined as a vertex of building exterior shape with an angle between adjacent line segments $\leq$ 170 degrees. It uses only external shape (`shapely.geometry.exterior`), courtyards are not included. Character is adapted from (Steiniger et al. 2008) to exclude non-corner-like vertices.

6. **Squareness of a building** is denoted as

$$Squ_{blg} = \frac{\sum_{i=1}^{n} D_{c_{blg_i}}}{n} \qquad (6)$$

where D is the deviation of angle of corner c_{blg_i} from 90 degrees and n is a number of corners.

7. **Equivalent rectangular index of a building** is denoted as

$$ERI_{blg} = \sqrt{\frac{a_{blg}}{a_{blgB}}} \times \frac{p_{blgB}}{p_{blg}} \qquad (7)$$

where a_{blgB} is an area of a minimal rotated bounding rectangle of a building (MBR) footprint and p_{blgB} its perimeter of MBR. It is a measure of shape complexity identified by Basaraner and Cetinkaya (2017) as the shape characters with the best performance.

8. **Elongation of a building** is denoted as

$$Elo_{blg} = \frac{l_{blgB}}{w_{blgB}} \qquad (8)$$

where l_{blgB} is length of MBR and w_{blgB} is width of MBR. It captures the ratio of shorter to the longer dimension of MBR to indirectly capture the deviation of the shape from a square (Schirmer and Axhausen 2015).

9. **Longest axis length of a tessellation cell** is denoted as

$$LAL_{cell} = d_{cellC} \qquad (9)$$

where d_{cellC} is a diameter of the minimal circumscribed circle around the tessellation cell polygon. The axis itself does not have to be fully within the polygon. It could be seen as a proxy of plot depth for tessellation-based analysis.

10. **Area of a tessellation cell** is denoted as

$$a_{cell} \qquad (10)$$

and defined as an area covered by a tessellation cell footprint in m^2.

11. **Circular compactness of a tessellation cell** is denoted as

$$CCo_{cell} = \frac{a_{cell}}{a_{cellC}} \qquad (11)$$

where a_{cellC} is an area of minimal enclosing circle. It captures the relation of tessellation cell footprint shape to its minimal enclosing circle, illustrating the similarity of shape and circle.

12. **Equivalent rectangular index of a tessellation cell** is denoted as

$$ERI_{cell} = \sqrt{\frac{a_{cell}}{a_{cellB}}} \times \frac{p_{cellB}}{p_{cell}} \qquad (12)$$

where a_{cellB} is an area of the minimal rotated bounding rectangle of a tessellation cell (MBR) footprint and p_{cellB} its perimeter of MBR. It is a measure of shape complexity identified by Basaraner and Cetinkaya (2017) as a shape character of the best performance.

13. **Coverage area ratio of a tessellation cell** is denoted as

$$CAR_{cell} = \frac{a_{blg}}{a_{cell}} \qquad (13)$$

where a_{blg} is an area of a building and a_{cell} is an area of related tessellation cell (Schirmer and Axhausen 2015). Coverage area ratio (CAR) is one of the commonly used characters capturing *intensity* of development. However, the definitions vary based on the spatial unit.

14. **Length of a street segment** is denoted as

$$l_{edg} \qquad (14)$$

and defined as a length of a `LineString` geometry in metres (Dibble et al. 2015; Gil et al. 2012).

15. **Width of a street profile** is denoted as

$$w_{sp} = \frac{1}{n}\left(\sum_{i=1}^{n} w_i\right) \qquad (15)$$

where w_i is width of a street section i. The algorithm generates street sections every 3 meters alongside the street segment, and measures mean value. In the case of the open-ended street, 50 metres is used as a perception-based proximity limit (Araldi and Fusco 2019).

16. **Openness of a street profile** is denoted as

$$Ope_{sp} = 1 - \frac{\sum hit}{2\sum sec} \qquad (16)$$

where $\sum hit$ is a sum of section lines (left and right sides separately) intersecting buildings and $\sum sec$ total number of street sections. The algorithm generates street sections every 3 meters alongside the street segment.

17. **Width deviation of a street profile** is denoted as

$$wDev_{sp} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(w_i - w_{sp}\right)^2} \qquad (17)$$

where w_i is width of a street section i and w_{sp} is mean width. The algorithm generates street sections every 3 meters alongside the street segment.

18. **Linearity of a street segment** is denoted as

$$Lin_{edg} = \frac{l_{eucl}}{l_{edg}} \qquad (18)$$

where l_{eucl} is Euclidean distance between endpoints of a street segment and l_{edg} is a street segment length. It captures the deviation of a segment shape from a straight line. It is adapted from Araldi and Fusco (2019).

19. **Area covered by a street segment** is denoted as

$$a_{edg} = \sum_{i=1}^{n} a_{cell_i} \qquad (19)$$

where a_{cell_i} is an area of tessellation cell i belonging to the street segment. It captures the area which is likely served by each segment.

20. **Buildings per meter of a street segment** is denoted as

$$BpM_{edg} = \frac{\sum blg}{l_{edg}} \qquad (20)$$

where $\sum blg$ is a number of buildings belonging to a street segment and l_{edg} is a length of a street segment. It reflects the granularity of development along each segment.

21. **Area covered by a street node** is denoted as

$$a_{node} = \sum_{i=1}^{n} a_{cell_i} \qquad (21)$$

where a_{cell_i} is an area of tessellation cell i belonging to the street node. It captures the area which is likely served by each node.

22. **Shared walls ratio of adjacent buildings** is denoted as

$$SWR_{blg} = \frac{p_{blg_{shared}}}{p_{blg}} \qquad (22)$$

where $p_{blg_{shared}}$ is a length of a perimeter shared with adjacent buildings and p_{blg} is a perimeter of a building. It captures the amount of wall space facing the open space (Hamaina et al. 2012).

23. **Mean distance to neighbouring buildings** is denoted as

$$NDi_{blg} = \frac{1}{n}\sum_{i=1}^{n} d_{blg,blg_i} \qquad (23)$$

where d_{blg,blg_i} is a distance between building and building i on a neighbouring tessellation cell. It is adapted from Hijazi et al. (2016). It captures the average proximity to other buildings.

24. **Weighted neighbours of a tessellation cell** is denoted as

$$WNe_{cell} = \frac{\sum cell_n}{p_{cell}} \qquad (24)$$

where $\sum cell_n$ is a number of cell neighbours and p_{cell} is a perimeter of a cell. It reflects granularity of morphological tessellation.

25. **Area covered by neighbouring cells** is denoted as

$$a_{cell_n} = \sum_{i=1}^{n} a_{cell_i} \qquad (25)$$

where a_{cell_i} is area of tessellation cell i within topological distance 1. It captures the scale of morphological tessellation.

26. **Reached area by neighbouring segments** is denoted as

$$a_{edg_n} = \sum_{i=1}^{n} a_{edg_i} \qquad (26)$$

where a_{edg_i} is an area covered by a street segment i within topological distance 1. It captures an accessible area.

27. **Degree of a street node** is denoted as

$$deg_{node_i} = \sum_{j} e\,dg_{ij} \qquad (27)$$

where edg_{ij} is an edge of a street network between node i and node j. It reflects the basic degree centrality.

28. **Mean distance to neighbouring nodes from a street node** is denoted as

$$MDi_{node} = \frac{1}{n}\sum_{i=1}^{n} d_{node,node_i} \qquad (28)$$

where $d_{node,node_i}$ is a distance between node and node i within topological distance 1. It captures the average proximity to other nodes.

29. **Reached cells by neighbouring nodes** is denoted as

$$RC_{node_n} = \sum_{i=1}^{n} c\,ells_{node_i} \qquad (29)$$

where $cells_{node_i}$ is number of tessellation cells on node i within topological distance 1. It captures accessible granularity.

30. **Reached area by neighbouring nodes** is denoted as

$$a_{node_n} = \sum_{i=1}^{n} a_{node_i} \qquad (30)$$

where a_{node_i} is an area covered by a street node i within topological distance 1. It captures an accessible area.

31. **Number of courtyards of adjacent buildings** is denoted as

$$NCo_{blg_{adj}} \qquad (31)$$

where $NCo_{blg_{adj}}$ is a number of interior rings of a polygon composed of footprints of adjacent buildings (Schirmer and Axhausen 2015).

32. **Perimeter wall length of adjacent buildings** is denoted as

$$p_{blg_{adj}} \qquad (32)$$

where $p_{blg_{adj}}$ is a length of an exterior ring of a polygon composed of footprints of adjacent buildings.

33. **Mean inter-building distance between neighbouring buildings** is denoted as

$$IBD_{blg} = \frac{1}{n}\sum_{i=1}^{n} d_{blg,blg_i} \qquad (33)$$

where d_{blg,blg_i} is a distance between building and building i on a tessellation cell within topological distance 3. It is adapted from Caruso et al. (2017). It captures the average proximity between buildings.

34. **Building adjacency of neighbouring buildings** is denoted as

$$BuA_{blg} = \frac{\sum blg_{adj}}{\sum blg} \qquad (34)$$

where $\sum blg_{adj}$ is a number of joined built-up structures within topological distance three and $\sum blg$ is a number of buildings within topological distance 3. It is adapted from Vanderhaegen and Canters (2017).

35. **Weighted reached blocks of neighbouring tessellation cells** is denoted as

$$WRB_{cell} = \frac{\sum blk}{\sum_{i=1}^{n} a_{cell_i}} \qquad (35)$$

where $\sum blk$ is a number of blocks within topological distance three and a_{cell_i} is an area of tessellation cell i within topological distance three.

36. **Local meshedness of a street network** is denoted as

$$Mes_{node} = \frac{e - v + 1}{2v - 5} \qquad (36)$$

where e is a number of edges in a subgraph, and v is the number of nodes in a subgraph (Feliciotti 2018). A subgraph is defined as a network within topological distance five around a node.

37. **Mean segment length of a street network** is denoted as

$$MSL_{edg} = \frac{1}{n}\sum_{i=1}^{n} l_{edg_i} \qquad (37)$$

where l_{edg_i} is a length of a street segment i within a topological distance 3 around a segment.

38. **Cul-de-sac length of a street network** is denoted as

$$CDL_{node} = \sum_{i=1}^{n} l_{edg_i}, \text{ if } edg_i \text{ is cul-de-sac} \qquad (38)$$

where l_{edg_i} is a length of a street segment i within a topological distance 3 around a node.

39. **Reached cells by street network segments** is denoted as

$$RC_{edg} = \sum_{i=1}^{n} cells_{edg_i} \qquad (39)$$

where $cells_{edg_i}$ is number of tessellation cells on segment i within topological distance 3. It captures accessible granularity.

40. **Node density of a street network** is denoted as

$$D_{node} = \frac{\sum node}{\sum_{i=1}^{n} l_{edg_i}} \qquad (40)$$

where $\sum node$ is a number of nodes within a subgraph and l_{edg_i} is a length of a segment i within a subgraph. A subgraph is defined as a network within topological distance five around a node.

41. **Reached cells by street network nodes** is denoted as

$$RC_{node_{net}} = \sum_{i=1}^{n} cells_{node_i} \qquad (41)$$

where $cells_{node_i}$ is number of tessellation cells on node i within topological distance 3. It captures accessible granularity.

42. **Reached area by street network nodes** is denoted as

$$a_{node_{net}} = \sum_{i=1}^{n} a_{node_i} \qquad (42)$$

where a_{node_i} is an area covered by a street node i within topological distance 3. It captures an accessible area.

43. **Proportion of cul-de-sacs within a street network** is denoted as

$$pCD_{node} = \frac{\sum_{i=1}^{n} node_i, \text{ if } deg_{node_i} = 1}{\sum_{i=1}^{n} node_i} \qquad (43)$$

where $node_i$ is a node within topological distance five around a node. Adapted from (Boeing 2017).

44. **Proportion of 3-way intersections within a street network** is denoted as

$$p3W_{node} = \frac{\sum_{i=1}^{n} node_i, \text{ if } deg_{node_i} = 3}{\sum_{i=1}^{n} node_i} \qquad (44)$$

where $node_i$ is a node within topological distance five around a node. Adapted from (Boeing 2017).

45. **Proportion of 4-way intersections within a street network** is denoted as

$$p4W_{node} = \frac{\sum_{i=1}^{n} node_i, \text{ if } deg_{node_i} = 4}{\sum_{i=1}^{n} node_i} \qquad (45)$$

where $node_i$ is a node within topological distance five around a node. Adapted from (Boeing 2017).

46. **Weighted node density of a street network** is denoted as

$$wD_{node} = \frac{\sum_{i=1}^{n} deg_{node_i} - 1}{\sum_{i=1}^{n} l_{edg_i}} \qquad (46)$$

where deg_{node_i} is a degree of a node i within a subgraph and l_{edg_i} is a length of a segment i within a subgraph. A subgraph is defined as a network within topological distance five around a node.

47. **Local closeness centrality of a street network** is denoted as

$$lCC_{node} = \frac{n-1}{\sum_{v=1}^{n-1} d\,(v,u)} \qquad (47)$$

where $d(v,u)$ is the shortest-path distance between v and u, and n is the number of nodes within a subgraph. A subgraph is defined as a network within topological distance five around a node.

48. **Square clustering of a street network** is denoted as

$$sCl_{node} = \frac{\sum_{u=1}^{k_v}\sum_{w=u+1}^{k_v} q_v\,(u,w)}{\sum_{u=1}^{k_v}\sum_{w=u+1}^{k_v}[a_v(u,w)+q_v(u,w)]} \qquad (48)$$

where $q_v(u,w)$ are the number of common neighbours of u and w other than v (ie squares), and $a_v(u,w) = \big(k_u - (1+q_v(u,w)+\theta_{uv})\big)\big(k_w - (1+q_v(u,w)+\theta_{uw})\big)$, where $\theta_{uw} = 1$ if u and w are connected and 0 otherwise (Lind et al. 2005).

49. **Connected buildings count** is denoted as

$$c_{blg} \qquad (49)$$

and defined as number of buildings directly adjacent to the target building.

50. **Connected buildings area** is denoted as

$$a_{cblg} \qquad (50)$$

and defined as total area of all buildings directly adjacent to the target building.

51. **Connected buildings perimeter** is denoted as

$$p_{cblg} \qquad (51)$$

and defined as total perimeter of all buildings directly adjacent to the target building.

52. **Connected buildings elongation** is denoted as

$$mibElo_{cblg} = Elo(cblg) \qquad (52)$$

where $cblg$ are all buildings adjacent to the target building and Elo is the elongation formula defined previously.

53. **Connected buildings equivalent rectangular index** is denoted as

$$mibERI_{cblg} = ERI(cblg) \qquad (53)$$

where $cblg$ are all buildings adjacent to the target building and ERI is the equivalent rectangular index formula defined previously.

54. **Connected buildings circular compactness** is denoted as

$$mibCCo_{cblg} = CCo(cblg) \qquad (54)$$

where $cblg$ are all buildings adjacent to the target building and CCo is the circular compactness formula defined previously.

55. **Connected buildings longest axis length** is denoted as

$$mibLAL_{cblg} = LAL(cblg) \qquad (55)$$

where $cblg$ are all buildings adjacent to the target building and LAL is the longest axis length formula defined previously.

56. **Connected buildings facade ratio** is denoted as

$$mibFR_{cblg} = \frac{mibAre_{cblg}}{mibPer_{cblg}} \qquad (56)$$

where $cblg$ are all buildings adjacent to the target building and $mibAre$ and $mibPer$ are the formulas defined previously.

57. **Connected buildings square compactness** is denoted as

$$mibSCo_{cblg} = \left(\frac{4\sqrt{mibAre_{cblg}}}{mibPer_{cblg}}\right)^2 \qquad (57)$$

where $cblg$ are all buildings adjacent to the target building and $mibAre$ and $mibPer$ are the formulas defined previously.

58. **Deviation of building area in tessellation neighbourhood** is denoted as

$$micBAD_{cell} \qquad (58)$$

and is defined as the standard deviation in the areas of all buildings within tessellation cells, directly adjacent to the target tessellation cell.

59. **Deviation of building area in node-attached buildings** is denoted as

$$midBAD_{node} \qquad (59)$$

and is defined as the standard deviation in the areas of all buildings attached to the target node.

There are three additional indicator variables calculated per morphotope - “Likely Occupied Area”, “Area of the largest ten connected structures” and “Perimeter of the largest ten connected structures”.

The “Likely Occupied Area” variable aims to identify morphotopes that mainly contain industrial buildings which tend to be blocky and not built to optimise light exposure. This variable is required due to inconsistencies in cadastre building definitions. For example,

some buildings that represent modern apartment blocks are split into multiple touching polygons representing the physical parts of the building such as an entrance; or an administrative part such as an address. Other modern apartment blocks, often in the same city, are presented as a single polygon even though they may have the same number of physical units (entrances) or administrative addresses associated with them.

To calculate it, the facade ratio - area over perimeter - is measured for each group of adjacent buildings within the morphotope. The final value of the “Likely Occupied Area” is 1 if more than 40% of the built up area with no courtyards has a facade ratio of greater than four and elongation less than .9 and furthermore.

The “Area of the largest ten connected structures” is calculated by summing the area of the largest ten connected structures in the morphotope. It is a measure of the built up area within the morphotope and is used for the final clustering.

Similarly, the “Perimeter of the largest ten connected structures” is calculated by summing the perimeters of the largest ten connected structures in the morphotope. It is not directly used in the taxonomy construction - its function is to detect outliers based on faulty cadastre data.

11. Spatial Agglomerative Adaptive Aggregation (SA3)

Spatial Agglomerative Adaptive Aggregation (SA3) combines two disparate clustering approaches.

1. First, it generates a full Ward clustering tree based on differences in feature space, and adjacency in geographic space.
2. Second it uses Leaf extraction to generate a set of clusters from the dendrogram.

The linkage matrix is generated by computing distances between observations based on the Ward formula, subject to a restriction that new connections must be spatially adjacent enclosed tessellation cells (ETCs). The leaf extraction algorithm processes the resulting dendrogram as follows:

1. The dendrogram is cut at all possible levels - one for each connection - starting from the lowest to the highest distance value.
2. At every level the number of members within each cluster and its constituent children are counted. If a cluster has more than N ETCs it is marked for extraction.
3. When one cluster marked for extraction merges with another, both are extracted from the dendrogram as separate clusters. Since the members of a marked cluster keeps increasing until a merger occurs, typically each extracted cluster has more than N members.
4. All points that are never part of a marked cluster are treated as outliers and marked as noise.

Before applying the clustering algorithm, all variables are preprocessed using a Quantile Transformer with a uniform distribution. This data transformation produces a relatively more equal weighing of all features when calculating distances between observations, as opposed to standardisation or normalisation. For example, variables with large ranges such as area, dominate less the distance calculations in feature space.

12. Data Normalisation & Outlier exclusion

Before the taxonomic tree is constructed, the data features have to be normalised to ensure individual variables do not dominate the distance calculations. We therefore drop any outliers that might affect this transformation. The need to do this arises again from the heterogeneous data quality in the cadastres of different countries. In some cases entire industrial areas are delineated as single buildings, in others power lines or sheds are present in the cadastre.

Specifically, we re-assign as noise any morphotopes that have either - a “Perimeter of the largest ten connected structures” larger than 200 km.; or “Area of the largest ten connected structures” larger than 500 sq.m.; or median building area less than 20 sq.m; or median building perimeter larger than 5km.

Once the outliers are re-assigned to noise, the rest of the data features are standardised. The outliers are processed together with the rest of the noise ETCs in subsequent steps.

13. Data preprocessing

13.1 Building preprocessing pipeline

- The first step in the is to split up multi-polygons and make the geometries valid.
- The second step, is to simplify the polygons in order to accurately represent the corners of buildings and other shape related characteristics.
- Next, to filter out any buildings that have an area larger than 200,000 sq.m. This is done since some artefacts such as construction sites, mines or tunnels might be included in the data as buildings.
- The next step is to merge overlapping buildings that either: overlap for at least 10 percent of their areas, or one of them has less than 50 sq.m. in total area. This is done to merge buildings and building parts, since cadastre definitions of these two polygon types are inconsistent and sometimes buildings are assigned as building parts or vice versa. This step merges the buildings and its parts into one polygon.
- Finally, the preprocessing pipeline snaps nearby buildings together and fills gaps in the polygons that are less than 10 square cm. These two steps aim to address some common topological issues, such as missing slivers with almost zero areas between multiple or inside individual building polygons. Nevertheless, even after

the preprocessing numerous topological issues remain and therefore we take this into account in subsequent analysis steps.

13.2 Street segments preprocessing pipeline

The types of street segments included in the analysis
living_street
motorway
motorway_link
pedestrian
primary
primary_link
residential
secondary
secondary_link
tertiary
tertiary_link
trunk
trunk_link
unclassified

The second major stage of the street processing is the simplification of the street network. The data from OvertureMaps, focuses on representation of street network for transportation purposes. That means it tends to include multiple geometries for wide boulevards where each captures a single carriageway, complex representation of junctions or even the smallest artefacts of transportation-based focus. However, such a network is not directly usable for morphological analysis as it does not capture morphological perception of street network which is usually captured via street centrelines, omitting transportation detail. For this reason, we apply the simplification method based on the detection of the problematic parts of the network (Fleischmann and Vybornova 2024). This ensures automatised algorithmic cleaning of street networks resulting in a morphological representation derived from the transportation one - dual carriage ways are merged together, roundabouts are simplified, and other changes.

13.3 Cadastral Buildings

Even though the data for each country comes from their respective official cadastre, there are still issues present.

First, there are cases where official cadastre data on building polygons within the same country comes from various sources, is in different formats and has inconsistent definitions of what a building is and how it should be split into units. Therefore, the data for

each country or region is downloaded individually and converted to the same format and only two-dimensional building polygons and no other properties are used for the analysis.

The inconsistent delineation issue is partially addressed through the design and selection of morphometric attributes. An extra step, specific only to Czechia is carried out to further address this. The boundaries of modernist housing estates in the official cadastre data are dictated by the coverage of local housing associations and not physical features. As such there is no consistency in their delineation across the country. To remedy this to some extent, we use a dataset on the spatial positions of communist era large scale developments and merge together the polygons that fall within them for consistency.

As a last step, we approximate building adjacency by assuming that any building within 50cm of another building is touching it and carry out all character calculations based on this assumption. This is done to avoid other subtle topological and numerical issues when dealing with high precision data.

13.4 Overture Maps street network data

The street network is a direct download from Overture Maps Transportation theme, a processed subset of data from OpenStreetMap, which has global coverage and high quality data. It is processed in two steps.

First, since the dataset includes multiple segment types, including footpaths and service roads that represent parking lot spaces, we limit the analysis to the segment types specified above. Another type of segment that is filtered out are tunnels - the analysis strictly focuses on two dimensions and therefore undergrounds structures adversely affect the calculation of boundaries and characters.

The second major stage of the street processing is the simplification of the street network. The data from OvertureMaps, focuses on representation of street network for transportation purposes. That means it tends to include multiple geometries for wide boulevards where each captures a single carriageway, complex representation of junctions or even the smallest artefacts of transportation-based focus. This information is secondary for morphological analysis and can adversely affect the calculated street attributes, therefore the network is standardised.

14. Cluster pen portraits

14.1 Level 1

Descriptions and names for the first two branches (level 1)

Cluster name	Cluster Description
Incoherent Fabric	Incoherent fabric covers a wide morphological variety, with a common theme of partial or complete breakage of the traditional structural roles of streets, plots, and buildings. Common for modernist period, post-

Cluster name	Cluster Description
	modern, and industrial developments, this branch typically has a less connected street network and may showcase buildings facing open spaces and internal parts of blocks rather than streets. At the same time, it contains less defined village developments.
Coherent Fabric	In coherent fabric, all streets, plots, and buildings take their traditional structural roles in defining the spatial arrangement of the urban form. It is common for traditional European development with densely connected street networks and legible plot structure, facilitating direct relation between buildings and streets.

14.2 Level 2

Descriptions and names for the first four branches (level 2)

Cluster name	Cluster Description
Incoherent Large-scale Fabric	Incoherent large-scale fabric captures typically urban development composed of buildings larger than the average, that may or may not be far from each other, creating large open spaces. Streets tend to be of an utilitarian use, rather than a structural one, typical for modernist housing estates or industrial zones.
Incoherent Small-scale Fabric	Incoherent small-scale fabric is mostly non-urban development capturing various kinds of villages and small towns, which show high variation of morphological properties. Buildings tend to be smaller, but distances between them vary, as well as the relations between buildings and streets.
Coherent Interconnected Fabric	This cluster has very high built-up density and local street connectivity, with narrow and short streets. It is primarily characterised by a high count of connected buildings forming enclosed blocks with courtyards. Furthermore, the distances between the formed blocks is small.
Coherent Dense Fabric	Dense fabric captures morphology typical for urban residential areas with lower density, where blocks are defined by streets more than buildings. The street networks are well defined and connected with buildings being either adjacent (e.g. row houses) or disjoint (e.g. urban villas).

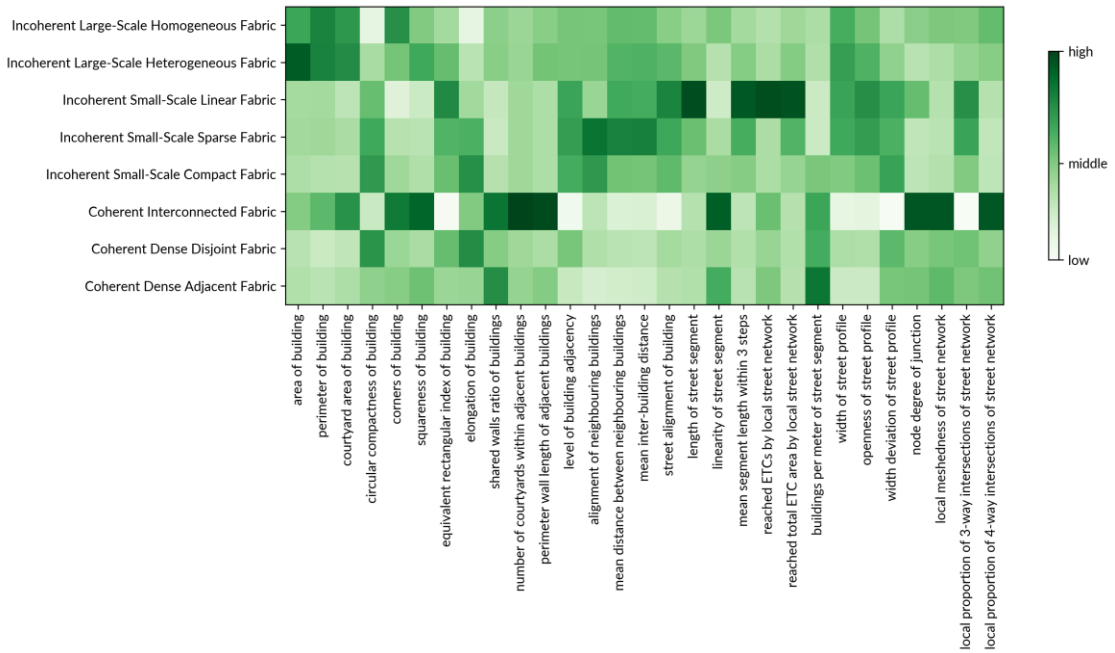
14.3 Level 3

Descriptions and names for the first eight branches (level 3)

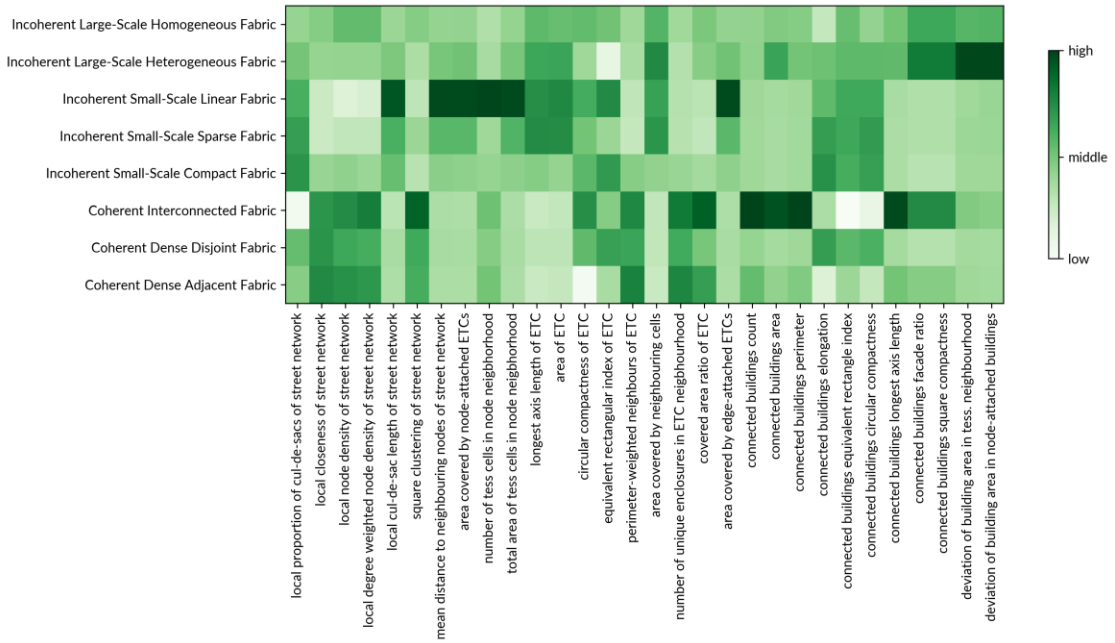
Cluster name	Cluster Description
Incoherent Large-scale Homogeneous	This cluster consists of the large buildings with moderate variations in size and shape, as well as low to moderate street connectivity and wide streets. The resulting environment is spacious, with

Cluster name	Cluster Description
Fabric	significant open areas between structures, typical of modernist housing.
Incoherent Large-scale Heterogeneous Fabric	This cluster consists of the largest buildings with notable variations in size and shape, as well as low to moderate street connectivity and wide streets. The design does not emphasize sunlight exposure, creating broad but less refined configurations, typical of industrial and other service areas.
Incoherent Small-scale Linear Fabric	This cluster has a moderate built up area and the low local street connectivity in the taxonomy, typically forming long linear villages. Its streets are long, linear, wide and there are minimal shared walls between structures.
Incoherent Small-scale Sparse Fabric	This cluster is characterized by low built-up density, low street connectivity, large distances between buildings, few shared walls, and large open spaces around buildings. The streets are few, open and wide. The buildings are small to moderate in size and their layout is more typical of rural areas.
Incoherent Small-scale Compact fabric	This cluster has low to moderate built-up area and street connectivity. Buildings exhibit a consistent alignment among themselves and also along streets of varying length, width and linearity. There is also a significant number of shared walls between structures.
Coherent Interconnected Fabric	This cluster has very high built-up density and local street connectivity, with narrow and short streets. It is primarily characterised by a high count of connected buildings forming enclosed blocks with courtyards. Furthermore, the distances between the formed blocks is small.
Coherent Dense Disjoint Fabric	This cluster has moderate to high built-up density and local street connectivity, with longer and wider streets compared to other dense developments. Shared walls between buildings are less common, and distances within buildings are moderate, reflecting a pattern of standalone structures within a robust street network .
Coherent Dense Adjacent Fabric	In this cluster, the built-up density and local street connectivity are high, while inter-building distances remain relatively small. Buildings frequently share walls, forming larger structures along relatively short and narrow streets.

14.4 Feature matrices of clusters



Part 1 of feature matrix showing the relative values of all morphometric attributes per each of the clusters at level 3.



Part 2 of feature matrix showing the relative values of all morphometric attributes per each of the clusters at level 3.

15. Adaptive limit for tessellation

Enclosed tessellation typically covers entirety of study area. In our case, we need to limit it only to those cells belonging to buildings while minimising the boundary effect on the edges of settlements generating unnaturally large cells. Hence, the delineation is further modified by introducing a variable, individual bandwidth for every building. The limits used here are calculated through a Gabriel graph-based filtering of the Voronoi graph of building centroids, which takes into account the surrounding neighbours structure around every building. This process results in a local bandwidth for every building based on distances to neighbouring buildings computed as the half of the maximum distance between neighbours of each building + 10% of the maximum distance. For example, in row housing the bandwidth will be relatively small, in single family housing estates the bandwidth will be larger, and in industrial areas larger even more; regardless of whether or not these buildings are in the middle of cities or around their edges.

16. Comparison

16.1 CORINE Land Cover classification

Figure 8 shows the distribution of the first three levels across CORINE Land Cover classes. In general, the branches of built-up fabric broadly follow the higher level patterns that are described by the CORINE Land Cover Classification. However, our branches split the CORINE classes into different sub-classes, providing more information on urban form, especially in more granular levels.



Figure 8: CORINE Land Cover cross-tabulation showing the proportion of each branch of built-up fabric within relevant CORINE classes.

"Discontinuous Urban Fabric" is split into multiple classes based on the density, building adjacency, street characteristics and overall distances between buildings by the clusters – *Incoherent Large-scale Homogeneous Fabric*, *Incoherent Small-Scale Compact Fabric*, *Coherent Dense Disjoint Fabric* and *Coherent Dense Adjacent Fabric*. It is worth noting

that both *Coherent Dense Disjoint Fabric* and *Coherent Dense Adjacent Fabric* are mostly discontinuous urban land, with *Coherent Interconnected Fabric* being nearly the only branch present within “Continuous Urban Fabric” class of CORINE. However, only half of it is “Continuous Urban Fabric”, indicating that the two concepts are not perfectly aligned.

The *Incoherent Small-Scale Sparse Fabric* and *Incoherent Small-Scale Linear Fabric* classes have the least amount of urban land and overlap with some agriculture and developed land classes. This is likely due to their presence within many rural areas. The *Incoherent Small-Scale Compact Fabric* stands in between these two more rural classes and the more urban ones. *Incoherent Large-scale Heterogeneous Fabric* overlaps the most with the CORINE’s Industrial and Commercial Units class, which is expected and highlights that heterogeneity at large scale in urban environments is typically serving non-residential purposes.

16.2 Copernicus Urban Atlas

Figure 9 shows a similar comparison with the Copernicus Urban Atlas. The results follow very similar patterns to the CORINE Land Cover comparisons, with some interesting differences due to the greater granularity of the Urban Atlas inside urban areas.



Figure 9: Copernicus Urban Atlas cross-tabulation showing the proportion of each branch of built-up fabric within relevant Urban Atlas classes.

Similarly to the CORINE comparison, *Incoherent Large-scale Heterogeneous Fabric* has a large overlap with industrial buildings. *Incoherent Large-scale Homogeneous Fabric* also intersects the most with this class, but 60% is split across the rest of the urban classes as these are often residential structures of typically modernist spatial organisation, but not

exclusively. *Incoherent Small-Scale Sparse Fabric* and *Incoherent Small-Scale Linear Fabric* overlap with the “Isolated structures” and lower density classes but none shows a dominance, reflecting a relatively weak alignment of their definition to the continuity-density definition used in Urban Atlas. *Incoherent Small-scale Compact Fabric* again stands in between these two more sparse classes and the more urban ones. In contrast to the CORINE Land Cover comparison, the *Coherent Interconnected Fabric* cluster is mainly falling in “Continuous Urban Fabric”, reflecting the fact that it covers mainly many interconnected buildings. *Coherent Dense Adjacent Fabric* is split between the Continuous and Discontinuous dense classes, in contrast to the previous comparison where it was mainly discontinuous. This indicates that not only are there differences between our classes and existing data products but also a degree of disagreement between the data products themselves.

16.3 Local Climate Zones table

Finally, Figure 10 illustrates the comparison with Local Climate Zones using the prediction by Demuzere et al. (2022). This broadly follows the same patterns as above. However, to put the comparison in context, around 47% of the Central European buildings are assigned to the “Open low-rise” LCZ class, with the second most common class “Sparsely built” having around 27% of the rest and the third one, “Low plants”, with 10%. Our most common classes are *Incoherent Small-Scale Compact Fabric* with 42%, *Coherent Dense Disjoint Fabric* and *Incoherent Small-Scale Sparse Fabric* with 15% each. Another thing to note is that there are a lot of buildings and fabrics that are classified in LCZ as part of the non-developed class, such as variations of plans and trees, which indicates the inability of LCZ to properly capture typically smaller settlements.

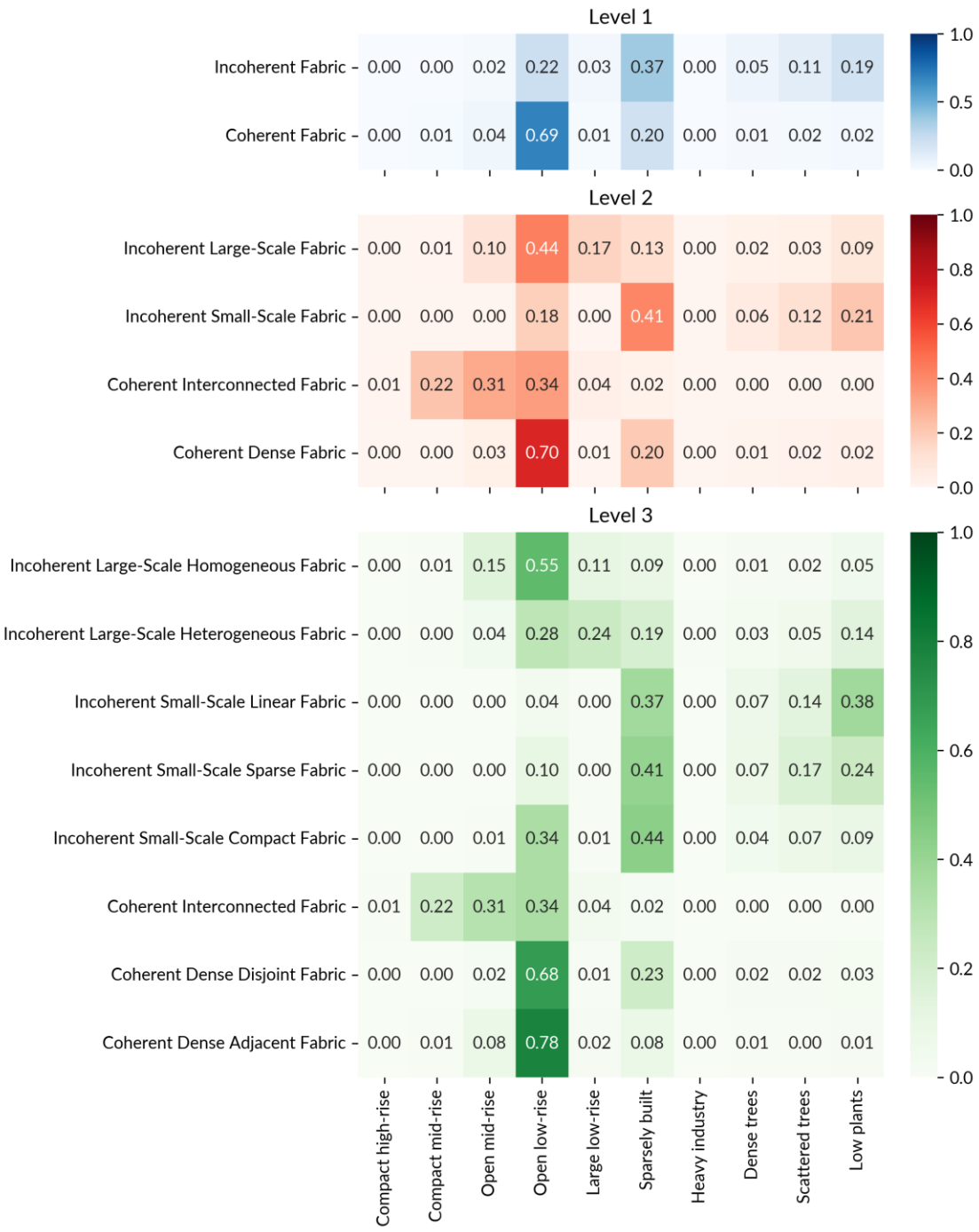


Figure 10: Local Climate Zones cross-tabulation showing the proportion of each branch of built-up fabric within relevant LCZ.

Similarly to the other comparisons, *Incoherent Large-scale Heterogeneous Fabric* has a large overlap with large low-rise buildings, representing the industrial areas. *Coherent Dense Disjoint Fabric*, *Coherent Dense Adjacent Fabric* and *Incoherent Large-scale Homogeneous Fabric* show the highest overlap with "Open low-rise", suggesting that we

are splitting this cluster based on characteristics that LCZ does not consider. *Coherent Interconnected Fabric* is the only cluster that has any overlap with compact mid-rise, but still is majorly open low-rise. *Incoherent Small-scale Sparse Fabric* and *Incoherent Small-scale Linear Fabric* overlap mostly with the "Sparsely built" cluster, as both cover primarily rural areas.

17. Scalability

The scalability of the method is evident from the extent of the study area, providing the classification for each of the 90 million buildings involved. However, what is important is that this has been achieved using a single high-end computer, rather than relying on a high-performance cluster or cloud or distributed processing. The specification of the machine used:

- 12th Gen Intel Core i7-12700 with 20 cores CPU
- 128GB RAM
- running Ubuntu 22.04

The estimated pure processing time, using a parallel processing pipeline on this single machine, is ~120 hours for the entire workflow excluding data retrieval (often heavily rate limited), from data pre-processing to generation of the data product. We believe that effective science needs to be computable on hardware accessible to many, minimising the barriers to adoption. While the machine we used is certainly high-end, it is still consumer electronics and even a machine with lower memory capacity and slower processor would be able to process the data.

18. Sensitivity of morphotope delination to morphometric attributes

To understand the relative importance of each of the 59 morphometric attributes that are driving the delination of morhotopes using the SA^3 algorithm, we have performed a sensitivity analysis using a leave-one-out technique. For a subset of the data (regions containing Prague, Vienna, Munich and Bratislava), we have delineated morphotopes 60 times: once with all of the attributes, and 59 times with one attribute always left out. The resulting delineation is then compared to the one from the full set of attributes using the adjusted mutual information (AMI) score REF. The lower values then indicate that leaving out that one particular attribute changes the outcome more than in cases with higher AMI.

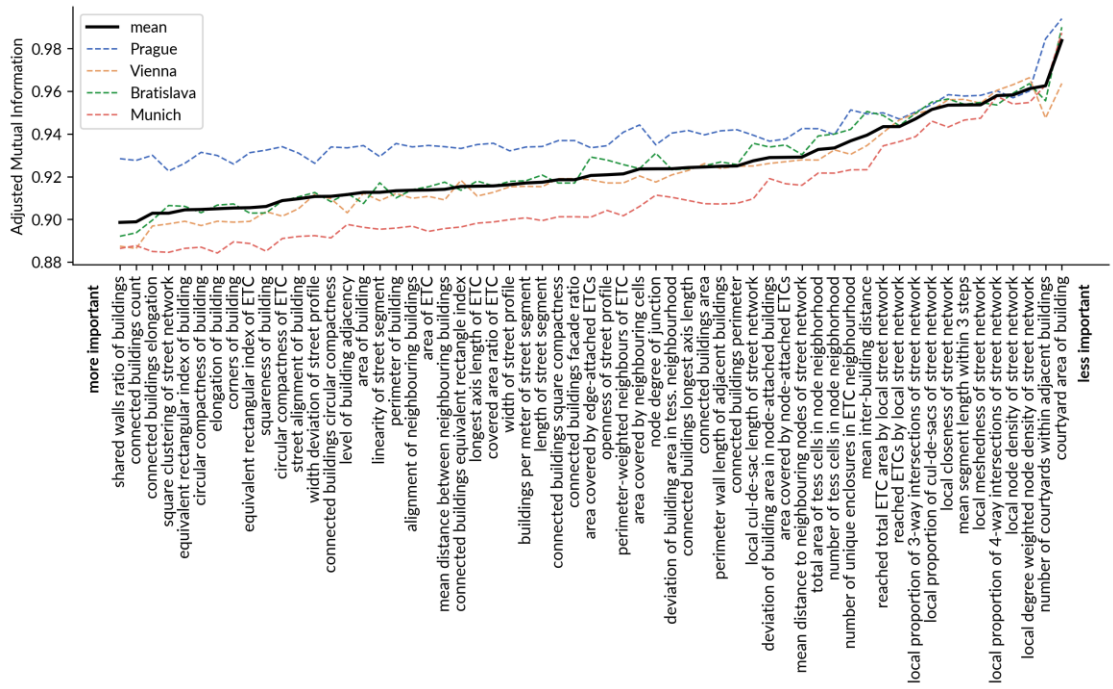


Figure 11: Relative importance of individual morphometric attributes in the morphotope delineation, sorted by mean AMI across 4 regions.

The results shown in Figure 11 highlight that the set is overall well balanced in terms of feature importance, with relatively minor differences in AMI between the delineations. An exception is *courtyard area of building*, which is less important than most of the others due to it being mostly zero in the vast majority of buildings. However, non-zero values tend to cluster in medieval urban fabric and the attribute plays a role in those circumstances. What the analysis also shows is that no single category of attributes is driving the delineation. Among the most important attributes (lower AMI), we observe those based on building distribution or shape as well as street network configuration or building-street alignment. However, what seems to play a generally lower role are network-based attributes, but the difference is not too large.

19. Optimisation of the parameter in morphotope delineation

The SA^3 regionalisation algorithm used for the delineation of morphotopes requires only a single parameter representing a minimal number of buildings within a region (i.e. morphotope). To determine the optimal value of this parameter, we analyse the range of options using the Total Within-Cluster Sum of Squares (TWSS), or inertia, as the primary objective function. Inertia measures the compactness of the resulting regions by quantifying the intra-regional variance.

For a set of K regions, inertia (I) is defined as:

$$I = \sum_{j=1}^{K} \sum_{i \in R_j} ||\mathbf{x}_i - \mu_j||^2 \qquad (60)$$

Where:

- R_j represents the set of spatial units assigned to region j.
- $\mathbf{x}_i$ is the feature vector of spatial unit i.
- μ_j is the centroid (mean vector) of region j.

Inertia acts as a proxy for information loss; it quantifies the heterogeneity introduced when distinct spatial units are forced to merge to satisfy the minimum size constraint (λ).

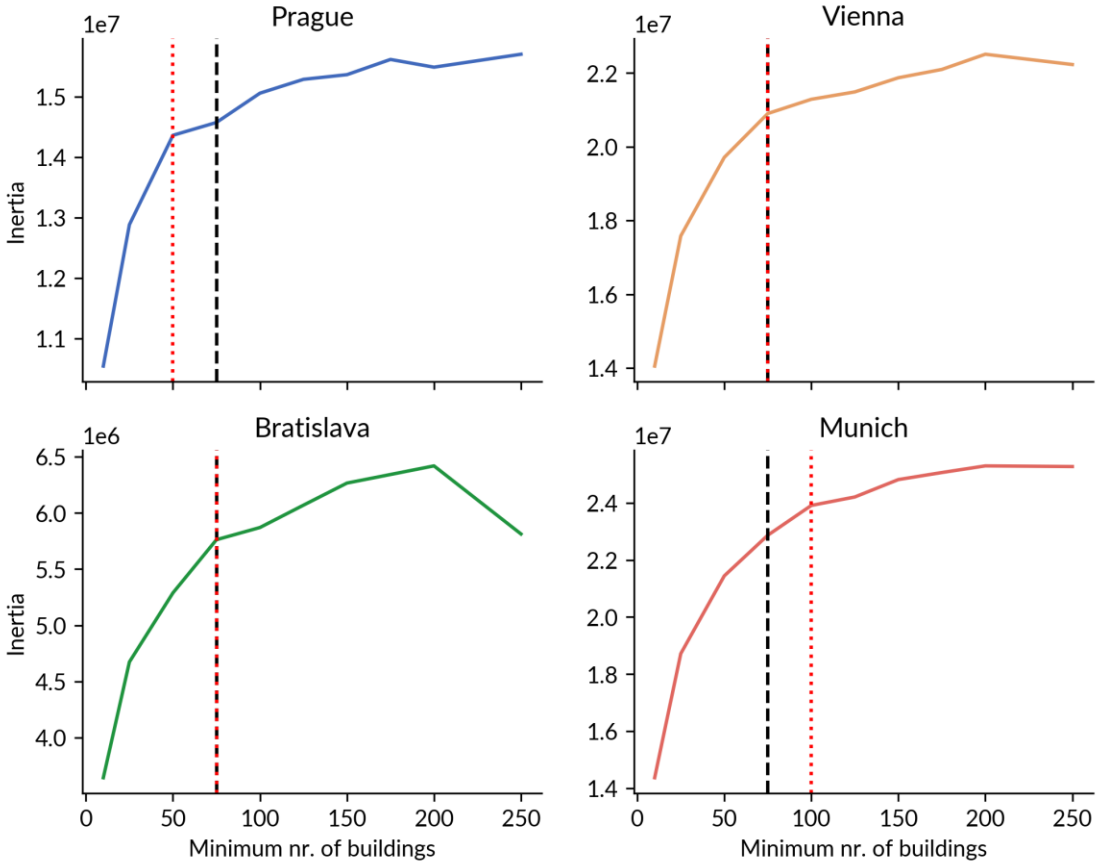


Figure 12: Inertia profiles for varying minimum number of buildings constraint across a sample of four regions from within the study area. Dashed vertical lines indicate the value 75, selected as the stabilisation point for the entire pipeline.

Figure 12 illustrates the change of inertia as a function of the minimum number of buildings (λ) across four regions of varying size and morphological composition. All curves exhibit a characteristic concave growth pattern with two distinct regimes.

First, we can observe a regime of rapid heterogeneity gain. In the interval between 0 and ~50-75, depending on the case, inertia increases steeply. This is likely due to inherent differences between individual buildings before they form a homogeneous pattern (morphotope). This underlines the notion of a morphotope as a pattern or a texture, not a uniform block. In this first interval, the algorithm is not yet forming morphotopes but suppressing local noise by forcing slightly different neighbours to share a mean value. Due to the high number of these merges, the inertia grows quickly.

Second, we can observe stabilisation of inertia. As (λ) increases beyond the inflection point, the rate of inertia growth slows down. In this phase, the algorithm is no longer forming regions from scratch but is likely merging adjacent clusters that already possess internal homogeneity. This transition point signals that we have surpassed the scale of the morphotope and are beginning to aggregate morphotopes into larger morphological regions.

The decrease in inertia in Bratislava and Vienna between 200 and 250 is primarily driven by a higher proportion of buildings marked as noise, keeping only the largest and most homogeneous areas. Proportion of noise is growing with growing λ, making the behaviour of the curve and its interpretation more complex than in a case of standard clustering like K-Means.

20. Relative importance of morphometric attributes for splits in the hierarchy

Each split in the hierarchy is driven by a different subset of morphometric attributes. The pen portraits already indicate that some of them are dependent more on attributes related to size, while others are related to street connectivity and so on. Figure 13 shows the relative importance of each of the attributes within the first 1000 bifurcations. It shows the number of times an attribute was among the top 10 whose values differ the most among the two branches being created. The difference is calculated as a difference in means per branch on top of standardised data used in the hierarchical clustering itself. Multiple conclusions can be made based on this analysis.

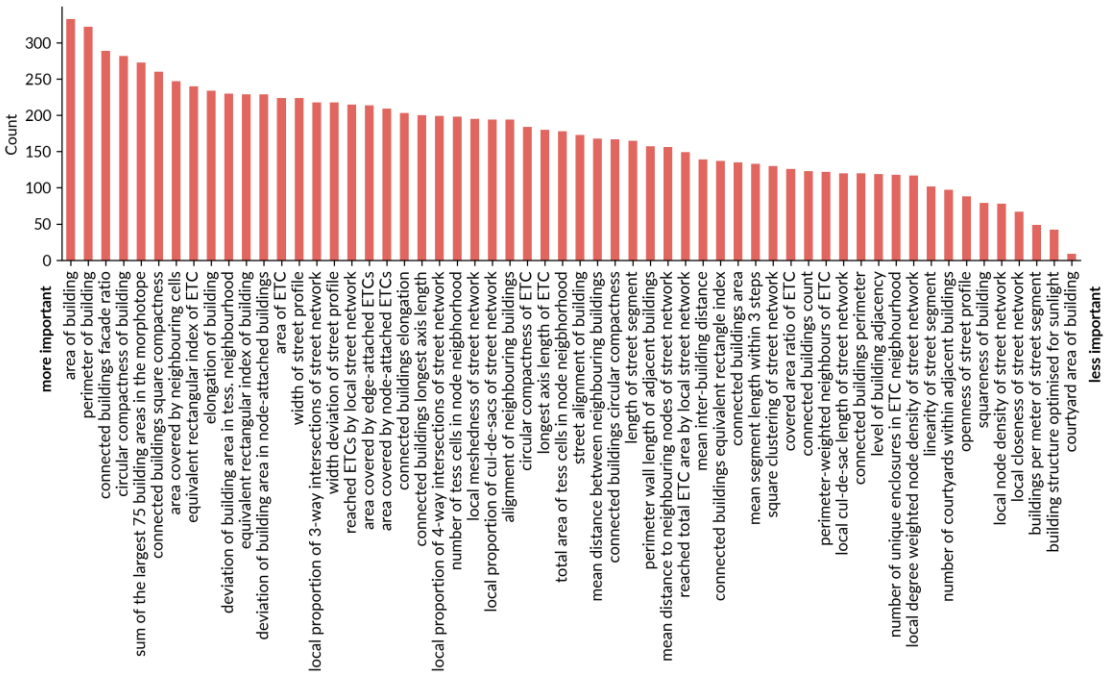


Figure 13: Relative importance of individual morphometric attributes for splits in the hierarchy, based on 1000 splits from the coarsest one. It shows the number of times each attribute belongs to the 10 with the highest distinction between the branches.

First, all of the attributes play their role. There is not a single one that is never among the top 10, despite the distribution not being uniform. The *courtyard area of building* plays the smallest role as in Section 18 but even that is important in 0.9% of the splits.

Second, the importance for the hierarchy differs from the importance for the morphotope delineation, with a subset of street network-related attributes that showed little importance for morphotopes driving ~20% of splits. This shows the necessity to design the set of attributes based on theoretical considerations rather than an empirical selection based on limited criteria. Each has its role in a different step of the classification pipeline and while some are less important in the beginning, their importance grows with the scale of the aggregation.

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